Teacher Wu Kuan | Ninth-Grade Exam at a Glance — Inscribed Angle = Half the Central Angle

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1. The trap question

Why is an inscribed angle exactly half of the central angle on the same arc?

2. The method

• Draw the diameter through the inscribed angle's vertex; two radii make isosceles triangles (base angles a and b).
• Exterior angle = sum of the two remote interior angles → central parts are 2a and 2b.
• Central = 2a + 2b, inscribed = a + b → exactly half.
• Consequences: angle in a semicircle = 90°; inscribed angles on the same arc are equal.

3. Parent check method

Ask your child to point to the arc both angles “look at” before applying the half rule.

Practice

1. A central angle is 100°. Find the inscribed angle on the same arc.
2. An inscribed angle is 35°. Find the central angle on the same arc.
3. AB is a diameter and C is on the circle. Find ∠ACB.
4. Two inscribed angles stand on the same arc. One is 40°. Find the other.
5. A quadrilateral is inscribed in a circle. One angle is 110°. Find the opposite angle.

Answers

1. 50°
2. 70°
3. 90°
4. 40°
5. 70°

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